



Andreas Garcia

Backend Engineer

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ABOUT

Backend engineer with a computer science engineering degree, I **design, ship and operate production APIs, microservices and data pipelines** in Python, Django REST Framework and **PostgreSQL**, deployed with **Docker** and **CI/CD** and monitored with Prometheus and Grafana. Implementing **62 accounting interfaces** for an **ERP** migration taught me to care about data consistency between systems, and I enjoy learning complex business domains quickly.

I founded **TheMusicTree**, where I am building the Django REST Framework backend of **GrowTheMusicTree**, whose staged bulk imports load **30.9M songs** and **6,000+ genres** from pipelines ingesting **MusicBrainz** and **Wikidata**. I run it myself, from Ansible provisioning to monitoring and backups. I also published **AudioMeta Python**, with 27k+ downloads.

SKILLS

- **Tech:** Python, Django REST Framework, **REST API**, PostgreSQL, Microservices, Docker, CI/CD, Grafana
- **Workflow:** Automated Testing, Code Review, Monitoring & Alerting, **Infrastructure as Code**, Specification
- **Languages:** French (native), **English** (advanced), Spanish (advanced), German (basic)
- **Hobbies:** WWOOFing, natural building (1 year), organizer of a music festival (80+ attendees)

EXPERIENCE

TheMusicTree - Founder & Developer

 **Dec. 2021 – Present**  **Paris, France**

- Building the Django REST Framework and **PostgreSQL backend** of **GrowTheMusicTree**, a world genre map whose staged bulk imports (Postgres COPY, set-based merge) load **30.9M songs** and **1.46M artists**.
- Building **TheMusicTreePipelines**, **Python/Polars ETL pipelines** ingesting **MusicBrainz** data and the live **Wikidata** SPARQL endpoint, with **data-quality checks** and a 90% coverage threshold enforced in CI.
- Operating the whole ecosystem with **Ansible** and **GitHub Actions** provisioning, **Prometheus** and **Grafana** monitoring, **Uptime Kuma** alerting, and offsite backups with an automated restore drill.

INRAE - Research Engineer

 **Oct. 2018 – Jan. 2020**  **Montpellier, France**

- Implemented event-handling **REST API** services in Java using the Data Transfer Object (DTO) pattern within **OpenSILEX**, an open-source research data platform.
- Manipulated complex graph data structures within the RDF4J database and improved ontologies for a **cross-disciplinary agrivoltaics research project** combining agriculture and photovoltaics.
- Collaborated within a **SCRUM** team on peer code reviews and contributed to making research data **FAIR** (Findable, Accessible, Interoperable, and Reusable) across the OpenSILEX codebase.

D. Cleversys → Deloitte - BI/ERP Consultant

 **Apr. 2015 – Oct. 2017**  **Paris, France**

- Specified and implemented **62 critical accounting interfaces** as **complex SQL scripts** for a large-scale ERP migration project, ensuring relational data integrity throughout the process.
- Served as **technical point of contact** for a client with annual revenue exceeding €4B, **leading specification workshops** and training developers on the ERP editor's ETL tools.
- Authored development guides and production deployment procedures for the ERP migration project's ETL tooling, while tracking and managing version releases of ETL processes.

EDUCATION

La Ruche - Programme Itinéraire (Entrepreneurship Program)

 **2025**  **Paris, France**

- Three-month entrepreneurship program with nine workshops (marketing, strategy, legal, etc.).
- Conducted market studies and developed a business plan.

INSA Lyon - Engineering degree in Computer Science

 **2014**  **Lyon, France**